1. Which computer is known as the first general-purpose electronic computer?
    A. UNIVAC
    B. ENIAC
    C. Colossus
    D. IBM Mainframe

2. Which of these is a core computer hardware component?
    A. Firewall
    B. Router
    C. CPU, RAM, Hard Disk, NIC
    D. Hypervisor

3. In which year did Amazon start Amazon Web Services (EC2)?
    A. 2004
    B. 2006
    C. 2008
    D. 2010

4. Which cloud service model gives virtual servers, storage, and networking?
    A. SaaS
    B. PaaS
    C. IaaS
    D. FaaS

5. Which are the important EC2 features?
    A. Static to Dynamic
    B. Distributed
    C. Auto Scaling, Load balancing, Elastic IP, EBS storage
    D. None of these

6. Which AWS service allows you to run code without managing servers (serverless)?
    A. Amazon EC2
    B. AWS Lambda
    C. Amazon EBS
    D. AWS Outposts

7. What is an Amazon Machine Image (AMI)?
    A. A physical hardware rack in a data center
    B. A software template used to launch an AWS EC2 instance
    C. A tool for monitoring network traffic
    D. A type of security firewall

8. Which service automatically adjusts the number of EC2 instances based on demand?
    A. Auto Scaling Group (ASG)
    B. Amazon EBS
    C. AWS cloudWatch
    D. Identity and Access Management (IAM)

9. Which Linux command is used to connect to the AWS EC2 instance?
    A. ssh -i@host IP address
    B. ssh -i ec2-user@host IP address
    C. ssh -i path/to/your-key-pair.pem user-name@EC2 instance public IP address
    D. ssh -chmod ec2-user@host IP address

10. Which security feature acts as a "virtual firewall" to control incoming and outgoing traffic to and from your EC2 instance?
    A. Key Pair
    B. IAM Role
    C. Security Group
    D. Target Group

11. An EC2 instance located in a private subnet requires access to objects stored in Amazon S3. Which solution should be implemented?
    A. Assign only an IAM role to the EC2 instance
    B. Provide internet access using an Internet Gateway
    C. Create a VPC Endpoint for S
    D. Assign a public IP address to the EC2 instance

12. Which AWS service transfers massive data (up to 100 PB) using a physical truck?
    A. Snowball
    B. Snowmobile
    C. Glacier
    D. DataSync

13. Which EBS volume type is recommended as the default due to independent scaling of IOPS and throughput?
    A. Gp2
    B. Gp3
    C. Io2
    D. sc1

14. Which statement about EFS storage is TRUE?
    A. You must define storage size before use
    B. It scales automatically without specifying size
    C. It supports only SMB protocol
    D. It is limited to a single EC2 instance

15. Which of the following statements about Amazon S3 is TRUE?
    A. Bucket names are unique only within a single AWS region
    B. A single S3 object can store unlimited data
    C. Data in an S3 bucket automatically moves across regions
    D. Bucket names must be globally unique across all AWS accounts.

16. A company wants a fixed public IP that can be remapped between EC2 instances. Which AWS feature should be used?
    A. NAT Gateway

B. Elastic IP
C. DHCP
D. Security Group

17. An application uses a protocol field in the IPv4 packet header. What does this 8-bit field define?
    A. The time to live in seconds
    B. The protocol used in the data portion of the IP datagram
    C. The size of the IP header in words
    D. The checksum of the Ethernet frame

18. A startup places its web servers in a subnet that can directly receive internet traffic. Which type of subnet is this?
    A. Hybrid Subnet
    B. Isolated Subnet
    C. Private Subnet
    D. Public Subnet

19. A cloud engineer configures a Network ACL (NACL) to block all inbound traffic on port 22 to secure a web server subnet. However, users are complaining that their SSH sessions are not working. Which of the following best explains this issue?
    A. NACLs are stateful, so you only need to allow inbound traffic.
    B. NACLs are stateless, and you forgot to allow the corresponding outbound traffic on the return ports.
    C. The engineer should use a Security Group instead of NACLs for the entire subnet.
    D. Port 22 is only used for HTTP traffic.

20. You want an EC2 instance in a private subnet to connect securely to an Amazon S3 bucket. You need to ensure that the data does not traverse the public internet. Which service should you configure?
    A. Internet Gateway
    B. NAT Gateway
    C. VPC Endpoint
    D. Peering Connection

21. An engineer accidentally applies a Terraform configuration that creates unwanted infrastructure. They now want Terraform to completely remove all managed resources from AWS. Which command should be used?
    A. terraform validate
    B. terraform init
    C. terraform destroy
    D. terraform graph

22. A developer configures CloudFront pricing incorrectly and repeatedly invalidates thousands of cached paths every day. At the end of the month, the AWS bill increases unexpectedly. Which CloudFront pricing factor MOST likely caused the increase? {
    A. Route table associations
    B. Invalidated cache paths beyond free limits
    C. Security Group ingress charges
    D. Availability Zone replication costs

23. An administrator configures a Security Group to allow inbound HTTPS traffic only. Users can successfully establish secure sessions, and return traffic works even though no explicit outbound HTTPS rule exists. Why does the communication succeed?
    A. HTTPS ignores outbound rules
    B. Security Groups are stateful
    C. NAT Gateway creates return sessions
    D. Route tables automatically allow HTTPS replies

24. A startup wants application traffic routed according to where users physically connect from. Canadian users should reach Toronto resources, while German users should reach Frankfurt resources regardless of latency measurements. Which Route 53 routing policy BEST satisfies this requirement?
    A. Geolocation routing
    B. Weighted routing
    C. Failover routing
    D. Simple routing

25. A company configures multiple healthy DNS records so Route 53 responds with up to eight random healthy endpoints during DNS resolution. Which routing policy is being used?
    A. Multivalue answer routing
    B. Geoproximity routing
    C. Latency routing
    D. Failover routing

26. An administrator configures a Security Group to allow inbound HTTPS traffic only. Users can successfully establish secure sessions, and return traffic works even though no explicit outbound HTTPS rule exists. Why does the communication succeed?
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    C. NAT Gateway creates return sessions
    D. Route tables automatically allow HTTPS replies

27. What is the primary operational trade-off when moving from a Relational Database (SQL) to Amazon DynamoDB (NoSQL)?
    A. You lose horizontal scalability.
    B. You lose microsecond in-memory performance.
    C. You trade rigid predefined schemas and strong joins for horizontal scaling.
    D. You lose the ability to define unique primary keys.

28. An application requires a cache that supports complex data models (like sorted sets) and must survive a node failure without data loss. Which engine should be chosen?
    A. A Memcached, because it supports multi-threaded persistence.
    B. Redis, because it provides data structures and supports data replication/snapshots.
    C. Memcached, because it shards data across multiple nodes for durability.
    D. Redis, because it natively runs SQL queries to reconstruct tables on crash.
29. An application requires a caching layer using Amazon ElastiCache. Which of the following architectural capabilities are unique to the Redis engine compared to Memcached?
    A. Native support for persistence, replication, and recovery features across cache nodes.

B. Extremely low-latency in-memory reads and writes for frequently accessed data.
C. Support for sophisticated data structures such as hashes, sorted sets, and pub/sub messaging.
D. Automatic horizontal scaling through stateless distributed key partitioning only.

30. An enterprise is migrating a legacy database to Amazon RDS. Which of the following statements accurately describe the operational and architectural benefits they will achieve by utilizing RDS?
    A. It offloads routine database tasks such as provisioning, patching, and hardware setup.
    B. None of the above.
    C. It offers synchronous Multi-AZ deployments for high availability and automatic failover.
    D. All the above statements are correct.

31. RDS1 is not configured with auto-scaling, and it ran out of storage space in the last hour. What is your recommendation?
    A. Add more space to RDS1 using the ModifyDBInstance option. Then enable storage auto-scaling thereafter.
    B. Create a new RDS2 with more storage space. Import it from the latest backup.
    C. Enable storage auto-scaling for RDS1 immediately.
    D. Configure a new read replica of RDS1 with more storage space.

32. What filesystem would you recommend for the EMR cluster?
    A. HDFS with a consistent view
    B. EMRFS with a consistent view
    C. HDFS
    D. EBS on the cluster node

33. A Spark step will be added to the EMR cluster for machine learning model training. What instances will you choose?
    A. C instance types
    B. R instance types
    C. T None of the option is correct
    D. All option is correct

34. Which Docker feature is most useful for ensuring a data analytics application runs consistently across development, testing, and production environments?
    A. Docker Hub
    B. Docker Containers
    C. Docker Compose
    D. Docker Swarm

35. A data engineering team uses Docker Compose for a pipeline containing PostgreSQL, Apache Spark, and Jupyter Notebook services. What is the biggest operational advantage of this setup?
    A. Compose enables direct kernel-level integration between Spark and PostgreSQL
    B. Compose guarantees zero-latency communication between containers
    C. Compose provides reproducible multi-container environments with service isolation and dependency management.
    D. Compose replaces Kubernetes for large-scale distributed analytics systems

36. In a Docker-based data analytics workflow, what is the primary purpose of a Docker Volume?

A. To increase CPU performance
B. To store persistent data outside the container
C. To create multiple containers automatically
D. To encrypt analytics datasets

37. Which AWS services are NLP-related services?
    A. Amazon transcript
    B. Amazon Comprehend
    C. Amazon terafit
    D. Amazon S3

38. Which AWS services are used for training the ML model?
    A. SageMaker
    B. Data Wrangler
    C. AWS Glue
    D. AWS EMR

39. A machine learning team has already prepared their data and trained several models. Their next priority is to evaluate whether predictions are systematically disadvantaging certain groups and to understand why the model is making particular predictions. Which SageMaker Studio tool is most appropriate?
    A. Model Explainability
    B. FairML
    C. Clarifies
    D. None of the option is correct

40. A company maintains a repository of reusable features that multiple teams can access to ensure consistency between training and inference environments. Which SageMaker Studio capability best supports this approach ?
    A. Data Wrangler
    B. Feature storage
    C. Feature store technique
    D. Autopilot

41. Which type of Auto Scaling policy is simplest to configure if you want to keep a specific metric, such as "Average CPU Utilization," at a constant value?
    A. Simple Scaling
    B. Step Scaling
    C. Scheduled Scaling
    D. Target Tracking Scaling

42. Which component of Elastic Load Balancing is responsible for performing health checks on registered EC2 instances?
    A. The Listener
    B. The Target Group
    C. The Scaling Policy
    D. The Security Group

43. The AZ naming example “us-east-1d” indicates
    A. Region us-east-1, AZ “d”
    B. Region us-east-d, AZ “1”
    C. Region us-1-east, AZ “d1”
    D. Region us-east, AZ “1d”

44. What were the main reasons CloudSpace failed to respond quickly to account compromises?
    A. AWS servers were permanently offline.
    B. IAM roles were deleted automatically.
    C. No threat-detection services were enabled.
    D. All option corrects

45. Which AWS services/logs does GuardDuty analyze to detect threats?
    A. Block DDoS service
    B. AWS CloudTrail.
    C. DNS logs.
    D. Amazon VPC Flow Logs.

46. A financial services company processes thousands of customer emails every hour. They need to:
        - Detect the dominant language of each email.
        - Identify person names, organizations, locations, and financial terms mentioned in the text
        - Minimize operational overhead and avoid managing ML infrastructure.

    Which AWS solution is the MOST appropriate?

    A. Use Amazon Translate for language detection and Amazon Rekognition for entity extraction.
    B. Use Amazon Comprehend to detect the language and extract named entities from the emails.
    C. Use Amazon Lex to detect the language and Amazon Textract to extract entities.
    D. Train a custom NLP model in Amazon SageMaker to perform language detection and entity recognition.

47. According to the AWS Shared Responsibility Model, which of the following is entirely the responsibility of AWS?
    A. Operating system and application patching
    B. Identity and access management configurations
    C. Physical security of data centers
    D. Data encryption at rest and in transit

48. What core security principle states that you should grant users only the minimum permissions required to perform their specific tasks?
    A. Separation of duties
    B. Defense in depth
    C. Least privilege
    D. Security by design

49. Which network component acts as a stateful firewall operating at the individual EC2 instance level?
    A. Network Access Control List (NACL)
    B. Security Group

C. Internet Gateway
D. NAT Gateway

50. Which AWS monitoring service uses machine learning to automatically detect intelligent threats, such as compromised EC2 instances or unauthorized access?
A. AWS CloudTrail
B. Amazon CloudWatch
C. Amazon GuardDuty
D. AWS Config